VIKRAM VARKOOR

AI & Hardware Engineer | Electrical Engineering Graduate | MSc AI (in progress) | Dubai, UAE
Email | GitHub | Portfolio | LinkedIn

Profile

Electrical engineer and MSc AI student working at the line between hardware and applied AI. Two published IEEE papers and a third under peer review. Shipped live AI applications (RAG, multi-agent LLM systems, a customer analytics warehouse) and embedded systems from PCB design and firmware to FPGA acceleration. Available now for full-time roles in AI, data and engineering.

Education

Heriot-Watt University Dubai Sep 2026 to present
MSc in Artificial Intelligence. Postgraduate Merit Scholarship and Duffin Family Future Shapers Scholarship. Semester 1: Artificial Intelligence and Intelligent Agents, Intelligent Robotics, Biologically Inspired Computation, Data Mining and Machine Learning.

American University in Dubai Graduated May 2026
BSc in Electrical Engineering. Coursework: Communication Systems, Digital Signal Processing, Systems and Controls, Microprocessors, Embedded Systems, Circuit Analysis, Electromagnetics.

Experience

Syncrow IoT: Hardware R&D Intern May 2025 to Sep 2025, ~422 hours, Dubai

- Validated and functionally tested 15+ IoT devices (Hikvision intercoms with facial recognition and fingerprint modes, radar presence sensors, smart locks, smart ACs, energy clamps, gateways) and catalogued specifications, accuracy and reliability for 30+ devices
- Designed a standardised 17-step device test procedure covering power-up, protocol handshake, connectivity, latency, backend data validation, UI correctness, edge cases and stress
- Ran regression testing on the Syncrow Analytics dashboard and wrote its validation requirements with the product owner and data scientist; worked with the Tuya platform and researched MQTT, ZigBee and wired versus wireless architectures

Research & Publications

- **IEEE ITEC-AP 2025, co-author:** Smart EV Charging Frameworks. Smart charging infrastructure, grid integration and optimisation.
- **IEEE ICTMOD 2024, co-author:** Microprocessor and ML Integration. Microprocessor architectures with ML pipelines for embedded and edge applications.
- **Under peer review (IEEE):** FPGA-Accelerated Edge AI for Clinical Decision Support: Architecture, Implementation and Performance Analysis.

Projects: AI, Data & Software

Churn-CLV-DBT | Customer analytics warehouse: leakage-safe churn, segmentation and CLV
Python, dbt (dbt-duckdb), DuckDB, LightGBM, PyMC-Marketing, pytest, Groq API | GitHub

- Built a parameterised dbt and DuckDB warehouse over 1,067,371 invoice lines and 5,942 customers with three time cutoffs, after a data audit that found 22.77% of rows missing a customer ID and 34,335 duplicate rows
- Enforced leakage safety with checks implemented twice (dbt test and pytest) and proven by deliberately planting a leak; evaluated churn models out-of-time with bootstrap confidence intervals (PR-AUC 0.801 to 0.814) and reported that no model clearly beat a simple recency rule
- Segmented customers with RFM and k-means (16.46% of customers drive 68.1% of revenue), modelled lifetime value with BG/NBD and Gamma-Gamma, and added an LLM summary layer guarded by a deterministic number and claim verifier; wrote the design doc before the code and logged 7 real bugs

Pitch Angle Finder | Full-stack AI PR pitch generator
Python, FastAPI, Pydantic, Next.js, TypeScript, Groq API, pytest, Render, Vercel, Azure | GitHub | Live

- Shipped a FastAPI and Next.js app that generates PR pitch angles from a live LLM, with Pydantic-validated JSON output and a retry mechanism for incomplete and malformed responses, tested against the real retry path
- Deployed two independent instances (Render and Vercel, and Azure) with CI/CD; diagnosed a production outage caused by a retired model and four real Azure deployment failures down to their root causes

Deriv Trading Agent | Autonomous paper-trading system with statistical backtesting
TypeScript, Next.js, Redux Toolkit, Jest/RTL, Cypress, Python, Supabase, Groq API | GitHub | Live

- Built an EUR/USD paper-trading agent: live 5-minute data, LLM BUY/SELL/HOLD decisions with confidence scores, code-enforced risk rules (0.55 confidence floor, 2:1 reward-to-risk) and a live Supabase dashboard
- Found and fixed a data leakage bug behind a misleading 95.88% backtest; the honest logistic regression result was 52.51% directional accuracy against a 46.59% baseline, with Wilson confidence intervals
- Added Redux Toolkit state, a 61-test Jest and RTL suite, one component built test-first (TDD), a 9-assertion Cypress end-to-end suite and custom Webpack configuration

Smart Meter Analytics | Distributed versus single-node data pipeline benchmark
Python, PySpark, pandas, SQLite, NumPy, matplotlib | GitHub

- Generated 10.5M rows of simulated smart meter data (300 devices, one year, 1.5% injected anomalies) as partitioned Parquet and implemented four analytics queries in both PySpark and pandas/SQLite
- Benchmarked at 100k, 1M and 5M rows: pandas was 7 to 30x faster at 100k, Spark was 8 to 10x faster at 5M on windowed aggregations, and pandas could not run the full set in memory

PaperChat | RAG-powered document chatbot
Next.js, FastAPI, fastembed (ONNX), NumPy, Groq API, Vercel, Render | GitHub | Live

- Built a RAG app where users upload a PDF and chat with it, with semantic search, source citations and token streaming (~800 tokens/sec via Groq)
- Replaced PyTorch and ChromaDB with fastembed (ONNX) and NumPy cosine similarity, cutting memory from about 500 MB to under 80 MB to deploy on a free tier

Lumen | Multi-model AI aggregator
Next.js, TypeScript, Supabase, Groq API, Vercel | GitHub

- Ran Llama 3.3 70B, Qwen 3 32B and Kimi K2 in parallel with concurrent SSE streaming, plus a judge agent that scores agreement (Jaccard) and synthesises a reconciled answer; Supabase auth, history and shareable links

AI Audit Risk Analyzer | Multi-agent audit system
Python, CrewAI, LangChain, Gemini API, Streamlit

- Built a 4-agent sequential CrewAI pipeline (risk identification, internal controls, report writing, governance review) with LangChain document ingestion; the governance agent checks output for ungrounded claims, PII and tone before release

Retail Operations Analytics | SQL and dashboard
Python, pandas, SQLite, SQL window functions, Tableau Public

- Built a Python ETL into SQLite over about 10,000 retail orders with four SQL queries (including LAG() month-over-month growth and CTEs) and a Tableau Public dashboard

Carbon Emission & Sustainability Tracker | Odoo module
JavaScript, Python, Odoo, PostgreSQL

- Odoo module for CSR and sustainability management: KPI tracking with threshold alerts and scheduled evaluation jobs, a geospatial site dashboard, and supplier audit and compliance tracking with role-based access

Projects: Embedded, Hardware & FPGA

Synapse.PL (HEAD System) | Senior design project, AUD EECE 498/499
Vitis HLS, Vivado, Zynq-7020, Python, XGBoost, Next.js, TypeScript

- Built a hardware-accelerated explainable Alzheimer's detection system: MRI to diagnosis in 17 seconds on a $269 Zynq-7020 FPGA (PYNQ-Z2) versus $3,000 to $10,000 GPU-class alternatives; 3-person team, sole contributor for the GLCM texture pipeline (252 Haralick features per scan)
- Implemented the accelerator in Vitis HLS (C++): 1.32 s bilateral extraction at 100 MHz with 13% LUT and 22% DSP utilisation, and an ARM Cortex-A9 to FPGA interface over AXI4-Stream DMA
- Trained an XGBoost and Optuna cascade on 3,436 ADNI scans (AUC 0.903 Alzheimer's vs normal, 0.803 normal vs MCI, 0.779 stable vs converting MCI) and built a Next.js clinical dashboard with 3D MRI viewer, Grad-CAM overlays and PDF reports

Smart Bin | Autonomous waste segregation
Python, OpenCV, YOLOv8n (Ultralytics), ONNX, Raspberry Pi GPIO

- Built a waste-sorting bin that classifies camera frames with a YOLOv8n model exported to ONNX and drives stepper and servo motors through Raspberry Pi GPIO, running fully on-device with no cloud dependency

Smaller hardware and systems projects

- **Smart Grid Theft Detector:** Arduino firmware flags load spikes on the device and streams readings at 115,200 baud to a live Python dashboard with a critical-alert state.
- **Power Quality Spectral Analyzer:** Arduino captures 64-point bursts; Python FFT computes THD and plots time and frequency domains live.
- **RFID Access Control:** Arduino and MFRC522 over SPI with UID checks, servo gate, LED and buzzer feedback and an interrupt-driven gate counter, in C++.
- **USB RGB LED Controller PCB:** schematic to fabrication-ready Gerbers in EasyEDA and Altium, with triple MOSFET PWM channels and LDO regulation.
- **AI Hands-Free OS Controller:** OpenCV head-lean tracking controls macOS volume through a native AppleScript bridge that avoids accessibility permissions.
- **Environmental Intelligence Dashboard:** async aiohttp client pulls OpenWeatherMap air quality data and plots pollutants against WHO thresholds.
- **Soil Moisture Monitor:** Arduino capacitive sensor with ADC calibration to a 0 to 100% reading.

Certifications

- **IBM, Agentic AI with LangChain and LangGraph:** issued August 2026, 100% grade.
- **IBM, Develop Generative AI Applications:** issued August 2026, 100% grade. Part of the IBM Generative AI Applications specialization.
- **Schneider Electric University:** Electric Vehicles: Plugging into Smarter Energy Management; Power Factor and Harmonics; Battery Safety and Environmental Concerns (2025).

Competitions & Achievements

- Khalifa Sustainability Competition: top 5 out of 50 teams
- AUS Sharjah Tourism Innovation Competition: shortlisted (invited teams only)
- A.I. for Sustainability Hackathon, Sorbonne University Abu Dhabi (October 2025): certificate of achievement
- Ruya AI Hackathon (February 2026): built Sovereign Agent, a self-improving AI assistant
- TranscenD 2025 Amazon Case Competition and AUD STEM Competition: participant

Skills & Technologies

- **Languages:** Python, TypeScript, JavaScript, C++, SQL, MATLAB
- **AI & ML:** LLM integration (Anthropic, OpenAI, Groq, Gemini), CrewAI, LangChain, RAG, multi-agent systems, scikit-learn, XGBoost, LightGBM, YOLOv8, OpenCV, ONNX Runtime, PyMC-Marketing, out-of-time validation, bootstrap confidence intervals
- **Data:** dbt, DuckDB, PySpark, pandas, NumPy, SQLite, Parquet, Tableau, matplotlib, FFT/DSP
- **Web & Cloud:** Next.js, React, FastAPI, Supabase, Tailwind CSS, Redux Toolkit, Jest, Cypress, Vercel, Render, Azure, GitHub Actions
- **Embedded & Hardware:** FPGA (Vitis HLS, Vivado), Arduino, Raspberry Pi, ESP32, GPIO, SPI/I2C/UART, PCB design (EasyEDA, Altium)
- **IoT:** MQTT, ZigBee, Tuya platform, device validation and testing
- **Languages spoken:** English (fluent), Hindi (native), Kannada (fluent), Telugu (conversational)